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Educational choice and marketization in Hong Kong: the case of direct subsidy scheme schools

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Abstract Direct subsidy scheme (DSS) schools are a product of Hong Kong's market-oriented educational reform, mirroring global reform that champions parental choice and school marketization. Such schools have greater autonomy in matters of curricula, staffing, and student admission. Although advocates of the DSS credit it with increasing educational diversity and competition, little empirical is available to back up such claim. In this study, we use data from the Program for International Student Assessment to compare student achievements in DSS schools with those in traditional public schools. We find that while DSS students' test scores in math, reading, and science have improved significantly over time, though the variation is much greater. Changes in mean performance have been anchored by a substantial change in student composition. DSS schools have a higher proportion of students with high socioeconomic status than with medium and low socioeconomic status. DSS schools also amplify the effects that family background have on student achievement. These findings raise concerns that the DSS approach favors a small minority of students.

Keywords Marketization · Privatization · Hong Kong · Secondary schools · PISA

Introduction

Given the current global trend of privatization, education in many industrial-capitalist societies has experienced progressive marketization. The direct subsidy scheme (DSS), an initiative of such marketization in Hong Kong, was launched in 1991 without much public enthusiasm. It was reintroduced in 1999 as a vehicle for encouraging diversity and improving quality in education. The DSS approach gained significant momentum in the early 2000s, as it granted autonomy and flexibility in various areas to schools that joined the scheme. The DSS was designed to (a) create a market for educational services to nurture a private sector of schools and (b) promote competition among private schools while extending inter-school competition from the private to the public sector. The argument was that inter-school competition would provide students with more school choices and quality education. Put simply, the DSS was intended as a mechanism promoting effectiveness, flexibility, innovation, and choice using the market approach (Legislative Council Panel on Education 2000).

The DSS reform was not spared criticism, but it gained popular support among parents and educators (Tse 2008). However, it remains to be seen whether the DSS can live up to its set goal of being an effective strategy for promoting quality education, as its effectiveness also hinges on how it influences the existing class inequalities in education. Given its adoption of the 'user pays' principle,

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the DSS plays a role in educational inequality by favoring students of relatively advantaged social backgrounds (Luk 1990). More than a decade after the implementation of the DSS, the number of primary and secondary DSS schools has increased from 12 in the 1996/1997 school year to 74 in the 2013/2014 school year (Secretary of Education 2014). Despite this drastic increase in the provision of education under the DSS, no effort has been made to empirically evaluate its effect on social inequality in education. We undertake this task in this study. Using three waves of data from the Hong Kong segment of the Program for International Student Assessment (PISA), covering most of the last decade, we describe the DSS's influence on student performance and its role in cream skimming students across schools. We also discuss the implications of our findings for educational equity.

In the next section, we introduce the background of the DSS and present the context of Hong Kong. Then, we review international evidence on similar school reform initiatives. We situate our research questions in the broad literature on educational choice and market reform and describe our data and research design, followed by analysis and discussion.

Literature review

The origins of the DSS and the Hong Kong context

Before the DSS was introduced, there were three types of schools in Hong Kong—government, private, and aided—distinguished by their source of funding and operational authority. Government schools are completely funded and run by the government. Private schools are funded and run by their sponsoring bodies and do not receive any subsidies from the government. Although government school students are not required to pay tuition fees, private school students must pay the tuition fees set by their individual schools. Aided schools are run by their sponsoring bodies but fully subsidized by the government. The students at such schools are not required to pay tuition fees but may be asked to pay other, miscellaneous fees demanded by their schools. Aided schools are also closely regulated by the government schools and thus are considered public. Before the DSS was launched, most schools were government or aided schools, collectively labeled as traditional public schools (TPS). There were very few private schools.

The initial introduction of the DSS in 1991 was intended to replace unpopular 'bought-place schemes' in private schools, in which the government used public resources to reserve enrollment quotas in private schools for public access (Bray 1995; Tse 2008). During most of the 1970s and 1980s, private schools were regarded as weak institutions for student learning and parents were unhappy about their children attending private rather than public schools. DSS was not met with enthusiasm. By the 1996/1997 school year, only 12 schools had joined the scheme.

Things turned around in the 2000s after more favorable conditions were offered to DSS schools. According to Tse (2008), the new scheme included increased developmental grants to schools and students, flexibility in the teachers' mandatory provident fund,[1] freedom of choice regarding the medium of instruction, and permission for feeding primary schools to also convert to DSS. These conditions proved attractive to many schools and were welcomed by the general public. Some 40 schools joined the DSS since 2000, and the number of participating schools has continued to climb steadily.

At the operational level, the DSS is designed to fall between the public and private extremes of school governance. DSS schools enjoy much greater autonomy than TPS in their operations and in setting the amount of their tuition fees, similar to private schools, but they also receive a public subsidy from the government on a per-pupil basis provided they fulfill certain requirements (Education Bureau 2011).[2] DSS schools are also subject to government evaluation of their managerial processes, although this scrutiny is much lighter than that imposed on public schools.

Debates on marketization in Hong Kong

From a policy perspective, the Hong Kong government has embraced the DSS in light of new public management ideas. It has been argued that traditional public schools did not live up to the expectation of providing high-quality education. Mechanisms such as DSS schools would inject diversity into the education system (Legislative Council Panel on Education 2000). The logic behind DSS schools is twofold: Firstly, parents will be given more freedom to choose where to send their children. Secondly, certain schools will be given more autonomy in operation. These two movements aim to create an education market scenario where parents as consumers can choose from varying educational providers, and schools have to compete for students to secure educational funding. The inter-school

[1] A form of pension fund. Teachers and their employers, schools, are both required to contribute.

[2] Each DSS school is eligible for a full subsidy from the government in the form of a block grant calculated on the basis of the DSS unit subsidy rate—i.e., the average unit cost of an aided school place—and the number of enrolled students at the DSS school, if the fee level of the DSS school is no more than $2\frac{1}{3}$ times the DSS unit subsidy rate. Indeed, the annual school fees for primary school students range from \$10,500 to \$78,000, and for secondary school students in junior and senior forms range from \$700 to over \$65,000 and from \$1600 to over \$98,000, respectively. As a reference, the median household monthly income is about \$23,000 (Census and Statistics Department 2014).

competition thus will lead to the emergence of schools that will focus on innovation and flexibility to devise curricula, pedagogies, and assessments that do a better job of meeting students' needs. Those that achieve such excellence will be rewarded by the market by receiving more students, whereas those that do not would be penalized by the market by losing students. This process would eventually result in the provision of quality education (Mok and Welch 2002). From this perspective, the possibility of positive educational change is thus to be found outside of the public school system, with public schools becoming targets of reconstruction. In Hong Kong, private schools have been deemed the answer to meeting new public demand (Legislative Council Panel on Education 2000). Given that the private education sector is very small, the reintroduction of DSS schools in the 2000s brought in elements of private school operation to public institutions while maintaining some regulatory procedures. The Hong Kong government envisioned DSS as a pioneer of using market force and parental choice in providing high-quality education (Tse 2008).

The increasingly positive reception of the DSS would not be possible if the government were the scheme's sole supporter. In fact, parents, particularly those in the middle class, favor the DSS due to their dissatisfaction with the public system. One major reason for this dissatisfaction is the government policy that mandated Chinese as the medium of instruction in all public schools. Parents who valued English flocked to DSS schools, which were exempt from the regulation.

In contrast, critics noted early on the possibility that the DSS might bifurcate the education system (Luk 1990). They noted that many of the schools that joined the DSS were traditional elite schools that had already established themselves among parents and students. Joining the DSS freed these schools from government regulation, allowing them to set tuition fees at the level of their choice and create their own market (Tse 2008).[3]

A related concern is that competition in the education market might lead schools to compete for the best students (so-called cream skimming) rather than focusing on improving the quality of their instruction (Tse 2008). DSS schools can admit students across school districts, whereas public schools are bound by their districts. DSS schools with an established market image are free to choose the students who best fit the schools' standards, whereas public schools are responsible for educating all who are eligible. The economic benefits of DSS schools are also questionable. If improved quality is achieved by devoting more resources to DSS schools, is it really a triumph for the DSS?

International evidence

These points are perhaps logically valid but not empirically grounded. In fact, a search of the related literature turned up only two student theses probing the effectiveness of DSS schools in the scheme's early stages, and both studies were small in scale (Yung 2006).

Given that no previous research has specifically examined the performance of DSS schools, in this section, we draw on the international literature on market-oriented reform at the elementary and secondary levels which are designed under same principal. For critics, the introduction of competition, choice, and new forms of school financing and governance directly challenges the traditional public school system (Ladd 2003). In the past three decades, such reform initiatives have often been associated with the creation or expansion of a private education sector, which has in turn put associated values such as equity and democratic control at stake (Ladd 2003). International evidence typically focuses on two aspects: efficiency gains in raising student achievement and the cream skimming of students from one school type to another.

On the first issue, the results have been mixed at best (Rouse and Barrow 2009). In the USA, numerous studies on charter schools have shown that they do no better or worse than traditional public schools (Bifulco and Ladd 2006; Ni 2009; Zimmer et al. 2009). Voucher programs have been reported to show positive gains by some (Hoxby 2003), but such gains have mostly been attributable to particular population subgroups (Krueger and Zhu 2004). Outside the USA, Hsieh and Urquiola (2006) reported that Chile's nationwide choice program did not improve average test scores, repetition rates, or years of schooling. This finding echoes earlier studies of a voucher system in Chile (McEwan 2001; McEwan and Carnoy 2000). In Sweden, where independent schools were introduced as a form of choice, Daun (2003) found that although the average test scores in these schools were higher, there was considerable variation between schools. The differences in school performance between sectors were largely attributable to differential enrollment of students from various socioeconomic backgrounds (Daun 2003), a point further highlighted by other cross-country studies (Somers et al. 2004).

On the second issue of cream skimming, the international evidence has indicated intensified student stratification under policies aimed at educational choice. Studies in the USA (Ni 2012; Bifulco and Ladd 2007), New Zealand (Fiske and Ladd 2003), and Chile (McEwan and Carnoy 2000) have all shown the increased isolation of students

[3] It is true that some DSS schools charge very high tuition fees, but there are also low-fee DSS schools. Yung (2006) analyzed the fees charged by DSS schools and found that about one-third were high-fee schools, which created the image of luxury schools. However, no analysis has been conducted in recent years, and thus, the present scale of this phenomenon is not known.

from disadvantaged socioeconomic or racial groups when choice was introduced. It is worth noting that the link between the cream skimming of students and market-oriented reform hinges to a great extent on the policy design (Hoxby 2003), and therefore, generalized interpretations should be made cautiously. Overall, empirical research has tended to find that highly educated families are more likely to exercise their choice (Cullen et al. 2005; Bifulco and Ladd 2007). These aspects suggest that it is reasonable to expect student stratification based on parental education and income in a free-for-all system.

To summarize, international evidence suggests that market-oriented approaches are of limited effectiveness in raising student achievement, and that their benefits are heterogeneous among students. The introduction of a new category of schools and the provision of choices to parents may thus lead to the segregation of students by race, ethnicity, or socioeconomic status. Again, however, this issue has not been examined in Hong Kong, in relation to the DSS.

Research questions

In our study, we use student achievement scores as an indication of education quality. Standardized test scores are widely used in the research on education marketization and parental choice (Ladd 2003). If inter-school competition works according to its design, then one would expect to see higher performance across the board. Consequently, in regard to DSS schools, we ask:

1. Has the introduction of the DSS generated higher achievement across the broader student population and/or just for DSS schools?

Secondly, in light of international literature, similar reform initiatives have led to a highly stratified student body across different types of schools. Therefore, we ask:

2. Does the DSS increase cream skimming across schools?

Thirdly, if cream skimming indeed happens, then:

3. What is the implication of DSS schools on overall educational equality in Hong Kong?

The first question directly examines the core assumption of the DSS; namely, that introducing new school types and giving them more autonomy may lead to inter-school competition, effectively 'lifting all the boats' (Hoxby 2003). How has student performance in each type of school changed since the DSS's introduction? The second research question is prompted both by theoretical consideration of the stratification effect of school competition and by international empirical findings. We also examine whether the differential benefits of DSS schools have implications for educational equity.

Data and method

In this study, we use student-level data from the PISA, one of the largest international assessments, conducted by the Organization for Economic Cooperation and Development (OECD). It tests 15-year-old students in reading, mathematics, and science on a 3-year cycle. Hong Kong is an early participant in the PISA, since 2000, but because school type can only be identified in the data from 2006 onward, we use data from the 2006, 2009, and 2012 waves of the PISA.[4] This approach allows us to test the performance of DSS schools and its change over time.[5]

The PISA uses a two-stage stratified sampling scheme (OECD 2012). Within each cycle, the Hong Kong PISA team divides schools by type. In the first stage, schools are randomly selected within each type. In the second stage, at least 35 students are randomly selected within each school. The students are given a background questionnaire and a battery of tests on the three test subjects. The response rate has averaged above 93 %.

Comparing student achievement across waves is only possible if the variables involved are scaled appropriately. In our case, the DSS designation has remained the same since 2006. The test scores, according to the OECD, have been scaled in such a way as to have the same mean and standard deviation across cycles (OECD 2009, 2012, 2014). The scale is normalized at a mean of 500 and a standard deviation of 100 points. The OECD has made it clear that one can only compare student achievement in all three subjects from 2006 onward (OECD 2014). Another important variable, the socioeconomic index of students, has also been rescaled to reflect changes in trends (OECD 2014, p. 258). The sample is described in Table 1.

To compare the mean performance in different school types, we follow the technique discussed in the PISA data analysis manual (OECD 2009). We also specify a multiple regression model to test whether the change in school performance is related to change in student composition. The structural model is specified in Eq. 1.

$$\text{Achievement} = \alpha + \lambda\text{Student} + \zeta\text{DSS} + \gamma\text{Year} + \sigma\text{SES} + \beta\text{DSS} \times \text{SES} + \varepsilon \quad (1)$$

'Achievement' is the test score in the three test subjects. 'Student' is a vector including student-level variables such as the child's sex, grade, language spoken at home, time

[4] Another thing that precludes us from using earlier cycles is that those test scores were not equated in a way that enables cross-cycle comparison. See OECD (2009) for details.

[5] Note that the schools sampled in each wave were different, even if they were given the same school identification number. Thus, a school-level longitudinal analysis is not possible. We only compare different types of schools in this analysis.

Table 1 Sample description

	Number of school		Number of students		Sample statistics (%)		
	Sample	Total	Sample	Total	Female	Speak Cantonese at home	Native
2006–2007							
Government	11	36	372	33,717	52	89	57
Aided	127	382	4031	385,003	50	92	56
Private/Int'l	1	57	24	20,289	38	NA	67
DSS	7	55	218	41,766	46	81	60
2009–2010							
Government	10	32	318	30,146	48	96	63
Aided	126	370	4046	365,213	48	94	59
Private/Int'l	2	60	61	22,984	41	83	84
DSS	13	61	412	51,123	38	80	66
2012–2013							
Government	10	32	310	26,313	84	95	67
Aided	119	365	3812	321,538	48	92	63
Private/Int'l	3	61	44	21,951	48	81	80
DSS	16	61	504	48,985	34	87	74

Source: OECD (2014); Secretary of Education (2009, 2014); Secretary of Education and Manpower (2007)

spent on academic subjects, and immigration status. 'DSS' is a dummy variable indicating whether students attend DSS schools. We exclude private and international schools from the regression due to an insufficient sample size. Equation (1) includes time dummy (γ) to account for the fluctuation in student achievement by PISA cycles. Socioeconomic status (SES) is a continuous index variable measuring students' economic, social, and cultural status. Additionally, because we are interested in understanding how student performance changes over time and by SES level, we include an interaction term to test for this idea.

After adding the interaction term, the coefficient σ now represents the association between achievement and SES from students when DSS is 0. This is the SES test score effect for non-DSS students:

$$\alpha + \sigma \mathrm{SES}$$

Similarly, the SES test score effect for DSS students (DSS = 1) is:

$$\alpha + \zeta + (\sigma + \beta)\mathrm{SES}$$

Analysis

In Table 2, we present the mean student performance for four school types over time. Student performance in all four types of schools improved in 6 years.[6] The rate of change seems to have accelerated between 2009 and 2012 compared to the 2006–2009 period. For instance, in science, the average performance of DSS students improved 15 points (0.15 standard deviation on the international scale) from 2009 to 2006 and the improvement increased by 37 points in the 2012–2009 period. Although the students in the government and aided schools also seem to have performed increasingly well, those in the DSS schools clearly improved much faster. This simple comparison, though intuitive, did not take into consideration of other micro- or macrolevel factor, and it should be interpreted as a necessary condition for RQ1 to be true.

Given that government schools usually admit the highest performing students, it is not surprising to see their students outperform the rest. Thus, a comparison of DSS with aided schools might be more meaningful, as the majority of the former were converted from the latter. Although DSS students lagged behind those in aided schools in 2006, in 2012 they performed at the same level.

Additionally, we test whether the change in mean performance has statistical significance. We test changes in mean performance, and changes in performance gaps between DSS students and students from other types of schools. For simplicity, the table and methodology are not shown here, but are available by request. We find that student performance did not improve across the board in all school types. Students in DSS schools experienced the most significant improvement in achievement, while others stagnated. By doing so, DSS schools closed the gap with aided schools ($z = -2.28$ for math and $z = -2.13$ for science), but the performance gaps between DSS and government school students and between DSS and international school students did not change.

This finding leads us to ask what DSS schools are doing, and what aided schools are not doing, to cause the DSS to

[6] The only exception is science subject in international schools.

Table 2 Performance in three subjects by school type

School type/subject	2006		2009		2012		Change in 2006–2012 *z* score
	M	SD	*M*	SD	*M*	SD	
Government							
Reading	562.3	82.5	553.2	78.5	570.8	85.9	0.57
Math	574.6	101.9	586.6	90.6	596.6	93.6	1.18
Science	569.8	91.8	575.5	80.6	582.2	81.2	0.97
Aided or caput							
Reading	537.2	80.1	535.4	83.0	543.6	84.4	0.98
Math	549.6	91.6	554.5	94.3	557.9	95.3	1.91
Science	544.3	90.0	550.7	86.4	553.3	81.6	1.85
Private or Int'l							
Reading	478.3	83.3	492.1	93.7	505.1	75.8	1.31
Math	524.6	82.6	538.4	90.9	520.2	95.8	−0.17
Science	509.5	95.5	515.3	89.6	499.8	95.6	−0.28
DSS							
Reading	508.9	87.3	503.7	86.0	540.2	87.8	1.41
Math	508.4	90.7	531.2	101.8	566.9	99.6	2.68
Science	503.7	94.6	518.4	91.8	555.8	90.9	2.57

catch up. We offer one possible explanation related to student composition, as suggested by Somers et al. (2004). In Table 3, we break down student composition, as measured by socioeconomic level, by school type and year. Low-, medium-, and high-SES students are those who belong to the bottom, middle, and upper third of the socioeconomic distribution, respectively, within each wave of data. We find that the DSS and TPS student bodies underwent substantial changes over the 6-year time span.

We first compute the median SES in each of the four school types. Then, we locate the percentiles of these SES scores in the student SES distribution for that year. The percentiles are presented in the second column of Table 3, and they represent the relative standing of a typical student from each type of school in the general student population.

In 2006, a typical DSS student ranked in the 64th percentile among all 15-year-old Hong Kong students. More than half of DSS students were low- or medium-SES students (23 and 29 %, respectively). Over time, the median DSS students rose in the SES ranking as the proportion of low- and medium-SES students in DSS schools shrank. In 2012, the typical DSS students rose into the 79th percentile in the SES distribution, and low- and medium-SES students constituted about one-third of the DSS student population (12 and 22 %, respectively). Over the same period (2006–2012), the SES ranking of the students in government and aided schools fell, from the 61st to the 53th percentile in government schools and from the 48th to the 45th percentile in aided schools. Simultaneously, the proportion of high-SES students in these types of schools fell, while that of low- and medium-SES students increased moderately. These figures lead us to suspect that DSS schools cream-skimmed top-SES students from the public sector during this period, a phenomenon identified by researchers at the birth of the DSS (Luk 1990; Chan and Tan 2008).

Table 3 Student SES distribution by school type

School type	Percentile of median SES student	Low-SES students (%)	Medium-SES students (%)	High-SES students (%)
2006				
Government	61th	26	30	44
Aided	48th	35	34	30
Private/Int'l	93th	0	8	92
DSS	64th	23	29	48
2009				
Government	48th	36	30	34
Aided	48th	35	36	29
Private/Int'l	95th	0	4	96
DSS	74th	23	20	57
2012				
Government	53rd	33	31	36
Aided	45th	37	36	27
Private/Int'l	91st	2	7	90
DSS	79th	12	22	65

This finding logically leads to the following questions. Is the change in student composition related to the change in school performance? How do student performance change

Table 4 OLS regression

Variable	Reading		Math		Science	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
DSS	−26.21***	−20.58***	−30.65***	−24.45***	−29.52***	−26.69***
	(5.37)	(5.51)	(5.23)	(5.25)	(5.20)	(6.24)
Year (2006 as reference)						
2009	−2.13	−2.30	8.67**	8.46**	−3.25	−3.31
	(2.85)	(2.82)	(3.14)	(3.13)	(3.16)	(3.17)
2012	9.22**	8.77**	15.51***	14.99***	5.67+	5.46+
	(3.06)	(3.05)	(3.53)	(3.50)	(3.24)	(3.27)
Female	25.58***	25.69***	−19.98***	−19.86***	−6.55**	−6.52**
	(2.53)	(2.55)	(2.92)	(2.93)	(2.36)	(2.37)
Language at home (Cantonese as reference)						
Chinese dialects	7.09	7.06	0.24	0.21	2.93	2.92
	(4.93)	(4.91)	(5.32)	(5.27)	(4.96)	(4.95)
English	−48.59***	−49.93***	−52.29***	−53.76***	−48.87***	−49.30***
	(12.45)	(12.37)	(13.00)	(12.75)	(14.05)	(13.94)
Other	−53.59***	−51.99***	−63.92***	−62.15***	−67.40***	−66.60***
	(5.32)	(5.19)	(5.44)	(5.33)	(7.76)	(7.66)
Immigration status (native as reference)						
Second generation	15.77***	15.76***	12.12***	12.11***	12.44***	12.41***
	(1.77)	(1.77)	(2.18)	(2.17)	(2.36)	(2.35)
First generation	24.76***	24.70***	21.98***	21.92***	24.19***	24.16***
	(2.56)	(2.60)	(2.72)	(2.73)	(2.89)	(2.91)
Grade	33.80***	33.65***	37.54***	37.39***	33.20***	33.13***
	(1.27)	(1.24)	(1.48)	(1.42)	(1.74)	(1.74)
Time spent in class (min)	0.05***	0.05***	0.13***	0.13***	0.13***	0.13***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
SES	20.11***	18.96***	24.11***	22.85***	19.00***	18.42***
	(1.26)	(1.30)	(1.27)	(1.37)	(1.26)	(1.30)
DSS*SES		10.40**		11.47***		5.22
		(3.59)		(3.47)		(3.40)
Constant	543.26***	542.17***	561.29***	560.19***	556.84***	556.35***
	(5.15)	(5.24)	(4.84)	(4.84)	(3.13)	(3.11)
R^2	0.2053	0.2067	0.2138	0.2150	0.2645	0.2648
N	11,952	11,952	11,911	11,911	9753	9753

Standard errors in parentheses

+ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

by SES level in DSS and TPS? What might the implications of the change in the student body be for educational equity? To address these questions, we examine the relationship between changes in student SES and those in student achievement using multiple regression models. The results are shown in Table 4. For the sake of simplicity, we show only two models for each subject.[7] Model 1 shows the estimates for various main effects. In Model 2, we test the relationship between the changing student body and achievement in three subjects in different types of schools by including interaction terms.

The main effects of the DSS, year, and SES are significant in the two specifications shown above. Overall, DSS students perform significantly worse than their peers in TPS in all three subjects when we control for PISA cycles, student background, and SES. Given that our aim is to understand whether high-SES students gain additional

[7] We fit multiple models by adding interaction terms one at a time. We eventually exclude those terms that are not significant.

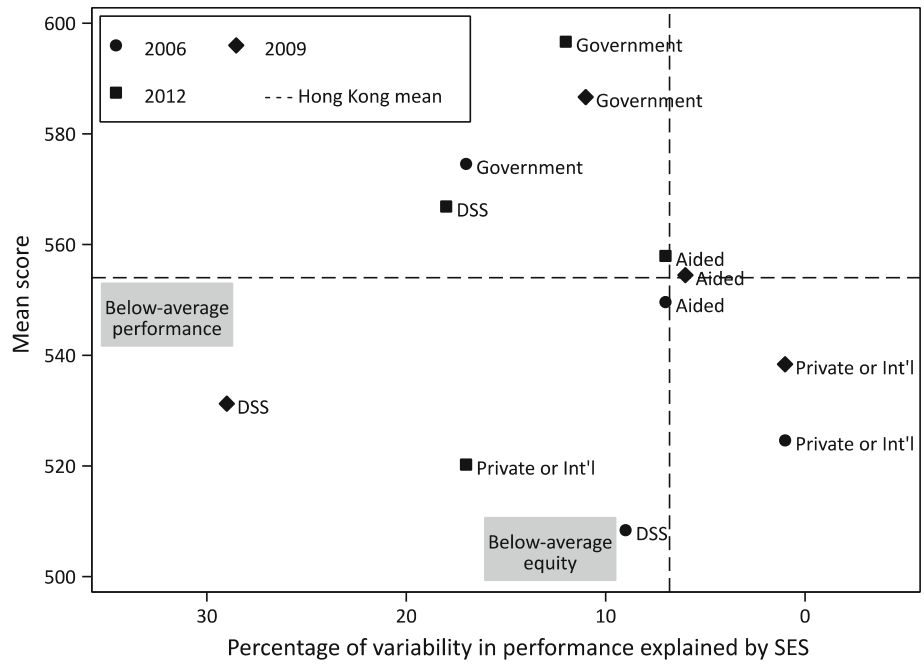

Fig. 1 Student performance in mathematics and equity by school type and year

advantage when they attend DSS schools, we pay close attention to the coefficient associated with DSS by the SES interaction term.[8]

Altogether, the relationship is more pronounced between student SES and achievement in DSS schools than in TPSs, as indicated by positive estimates of the interaction terms in reading and mathematics. The advantage of high-SES students enrolled in DSS schools over low-SES students is about one-tenth of the standard deviation. Thus, by enrolling more high-SES students, DSS schools reap performance gains and close the performance gap with aided schools.

If DSS schools do not benefit students evenly, what might the implications be for educational equity and performance? We present one analysis based on the methodology proposed by the OECD (OECD 2013). Figure 1 illustrates the relationship between mean performance in mathematics and a student's socioeconomic status, as measured by the percentage of variation in achievement scores explained by the SES variable (R^2). The four quadrants can be described as follows:

1. upper right quadrant: above-average performance and above-average equity;
2. upper left quadrant: above-average performance and below-average equity;
3. lower left quadrant: below-average performance and below-average equity; and
4. lower right quadrant: below-average performance and above-average equity.

[8] We also tried to control for school specificity by using a school fixed effect, but the number of schools was too low for the model to converge.

While Hong Kong is categorized under 'above-average mathematics performance' along with 'above-average equity in education outcomes' (OECD 2013, p. 27), our analysis reveals variation among school types. The students in the aided schools align most closely with the average equity-performance combination, as they recruit the most students. Yet for DSS students, a higher percentage of math performance can be explained by SES background than is the case for students in government or aided schools. Repeating this analysis using performance scores in reading and science yields qualitatively similar results, suggesting that the relationship between family background and achievement is stronger in DSS schools than in other types. Not only are DSS schools enrolling students selectively, but they are also probably reinforcing the family effect on learning outcomes.

Conclusion

Although the PISA data enabled us to capture changes in student performance in different school types over time, they did not permit us to tackle the issue of the endogeneity of choice options (Hoxby 2003). We were not able to follow schools across PISA cycles or to control for selection bias of student choosing different types of schools, nor did we have prior knowledge of the backgrounds of the students who attended DSS schools. This is the biggest limitation of our study. Because school administrators had varying reasons for choosing to convert their schools into DSS schools, it was possible that those schools that did join the DSS were very different from those that remained in

the public school sector. Thus, our second limitation is that our comparison probably does not reflect true differences between DSS and public schools but is confounded by unobserved school and student factors. In particular, what needs to be taken into account is the effect of students leaving TPS for DSS on achievement. In order to fully understand the effect DSS schools have on the entire education system in Hong Kong, researchers will need longitudinal data on schools that captures student performance before and after the introduction of DSS. PISA certainly does not provide such information, and our analysis is descriptive and correlational in nature rather than a causal evaluation of the effectiveness of DSS.

Regardless of these limitations, our study is the first to empirically question the prevailing paradigm in education and public policy discourse in Hong Kong. We show that overall achievement does increase in conjuncture with expanding DSS sectors (RQ 1) although the exact nature of this simultaneous occurrence still needs further examination. While the test scores of students in DSS schools improved, DSS schools did not seem to close the gap with government schools. The gap between DSS schools and aided schools did shrink, but this was largely due to substantial changes in DSS schools' student composition. Over the course of 6 years, the DSS student body has become increasingly weighted toward high-SES students. Ultimately, the DSS could arguably be seen as serving only the privileged portion of the population (RQ2). Once we accounted for students' socioeconomic background, students in DSS schools actually performed worse in all three assessed subjects.

The direction of change in student composition points to the third key finding; namely, that the socioeconomic effect on student learning outcomes is heightened in DSS schools compared to other types of public schools (RQ3). This suggests that DSS schools are more unequal than TPS due to the stronger SES net effect among DSS students. Considered within the context of DSS schools, this finding is not particularly surprising because many DSS schools promote extracurricular enrichment activities among their students (Yung 2006). Yet, only those with sufficient social capital can take advantage of such opportunities. Accordingly, our findings suggest that low-SES students enrolled in DSS schools did not gain as much as their counterparts in aided schools.

There is no denying that there are terrific schools under the DSS. This analysis is not an indictment of individual schools. Instead, it aims to ask the question whether the DSS as a general approach to school reform achieves what it claims to, and if so, at what cost.

When the DSS was first proposed, the research community warned that it could undermine the meritocratic tradition of Hong Kong society. The argument was that if DSS schools were exempted from the regulations imposed on public schools and were permitted to charge whatever fees they chose, they would end up serving only those who could afford the high fees and ultimately become another institution to benefit the already privileged (Luk 1990; Chan and Tan 2008). Yet this concern was quickly discarded when system-wide reforms championed the leading role of the market mechanism as key to improving education quality (Mok and Tan 2004). The discourse of diversity, competition, and excellence has become the epitome of school policy in Hong Kong, without being subjected to empirical scrutiny.

While this discourse remains appealing to many, we believe that it requires close examination, because the futures of thousands of students are at stake. If issues of economy, efficiency, and effectiveness are constantly scrutinized in public schools, why should semi-private schools, such as DSS schools, be let off the hook?

These findings we presented are certainly far from definitive; instead, they lead us to question the assumptions behind the DSS. Given DSS schools' fee structure and admission hurdles, we doubt whether they were ever intended to be a true alternative to their public counterparts. Savvy educational entrepreneurs see the DSS as an opportunity to break away from governmental regulation, but these schools are not aimed at educating the same children as public schools. The Hong Kong government once envisioned the DSS as a pioneer of private education before the full-scale conversion of public schools to the private sector (Yung 2006). Yet our findings suggest evidence that DSS schools only serve an increasingly narrow segment of the student population. Seemingly, the tidal wave of privatization has not lifted all of the boats; rather, it has created a separate stream open to some at the cost of others. The public schools that have lost high-SES students are probably the biggest losers in this market. The implication is such that hereditary privilege has resurfaced and belief in equity being has been squandered. This, to our view, is the hidden cost of the DSS approach to the general education system. We thus conclude by calling more systematic examinations of DSS schools, not just the quality of education but its effect on education inequality.

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